PC1 (2024/2025) - ANALYSIS I
EXERCISE SET 02 - COMPLETE SOLUTIONS

Exercise 1

Problem 1. Show that for any $x, y, u, v \in \mathbb{R}$:

(i) $|x| + |y| \leq |x + y| + |x - y|$

(ii) $|x - u| - |y - v| \leq |x - y| + |u - v|$

(iii) $\frac{|x+y|}{1+|x+y|} \leq \frac{|x|}{1+|x|} + \frac{|y|}{1+|y|}$

(iv) $1 + |xy - 1| \leq (1 + |x - 1|)(1 + |y - 1|)$

Solution. **(i)** Consider the following algebraic manipulation:

$$\begin{aligned}(|x+y| + |x-y|)^2 &= (x+y)^2 + (x-y)^2 + 2|(x+y)(x-y)| \\ &= 2x^2 + 2y^2 + 2|x^2 - y^2| \\ &\geq 2x^2 + 2y^2 \quad (\text{since } |x^2 - y^2| \geq 0) \\ &= (|x| + |y|)^2 + (|x| - |y|)^2 \\ &\geq (|x| + |y|)^2\end{aligned}$$

Taking square roots of both sides gives the desired inequality.

(ii) By the reverse triangle inequality and regular triangle inequality:

$$\begin{aligned}|x - u| &= |(x - y) + (y - v) + (v - u)| \\ &\leq |x - y| + |y - v| + |v - u| \\ &= |x - y| + |y - v| + |u - v|\end{aligned}$$

Thus:

$$|x - u| - |y - v| \leq |x - y| + |u - v|$$

(iii) Consider the function $f(t) = \frac{t}{1+t}$ for $t \geq 0$. Its derivative is:

$$f'(t) = \frac{1}{(1+t)^2} > 0$$

so f is strictly increasing. By the triangle inequality $|x + y| \leq |x| + |y|$, we have:

$$f(|x + y|) \leq f(|x| + |y|)$$

Moreover, f is subadditive :

$$\begin{aligned}f(a+b) = \frac{a+b}{1+a+b} &\leq \frac{a}{1+a} + \frac{b}{1+b} \\ &= f(a) + f(b)\end{aligned}$$

Thus:

$$\frac{|x+y|}{1+|x+y|} \le \frac{|x|+|y|}{1+|x|+|y|} \le \frac{|x|}{1+|x|} + \frac{|y|}{1+|y|}$$

(iv) Let $a = x-1$ and $b = y-1$. Then:

$$\begin{aligned} 1+|xy-1| &= 1+|(1+a)(1+b)-1| \\ &= 1+|a+b+ab| \\ &\le 1+|a|+|b|+|ab| \\ &= (1+|a|)(1+|b|) \\ &= (1+|x-1|)(1+|y-1|) \end{aligned}$$

Exercise 2

Problem 2. Let x, y be real numbers, show that:

(i) $\max(x,-x) = |x|$

(ii) $\max(x,y) = \frac{1}{2}(x+y+|x-y|)$

(iii) $\min(x,y) = \frac{1}{2}(x+y-|x-y|)$

(iv) $\max(|x|,|y|)\left|\frac{x}{|x|} - \frac{y}{|y|}\right| \le 2|x-y|$ for $x, y \neq 0$

Solution. (i) Direct computation:

$$\max(x,-x) = \begin{cases} x & \text{if } x \ge 0 \\ -x & \text{if } x < 0 \end{cases} = |x|$$

(ii) Case analysis:

- If $x \ge y$, then $|x-y| = x-y$, so:

$$\frac{1}{2}(x+y+x-y) = x = \max(x,y)$$

- If $y > x$, then $|x-y| = y-x$, so:

$$\frac{1}{2}(x+y+y-x) = y = \max(x,y)$$

(iii) Similar to (ii):

- If $x \ge y$, then:

$$\frac{1}{2}(x+y-(x-y)) = y = \min(x,y)$$

- If $y > x$, then:

$$\frac{1}{2}(x+y-(y-x)) = x = \min(x,y)$$

(iv) Without loss of generality, suppose $|x| \ge |y|$. Since $x, y \neq 0$, consider

$$\left|\frac{x}{|x|} - \frac{y}{|y|}\right| = \left|\frac{x|y| - y|x|}{|x||y|}\right| = \frac{||y|x - |x|y|}{|x||y|}.$$

Applying the triangle inequality to the numerator, we obtain

$$||y|x - |x|y| = ||y|(x-y) + y(|y|-|x|)| \le |y||x-y| + |y|\,||y|-|x||\,.$$

Since $||y|-|x|| \leq |x-y|$ by the reverse triangle inequality, it follows that

$$||y|x - |x|y| \leq |y||x-y| + |y||x-y| = 2|y||x-y|.$$

Substituting this into the earlier expression gives

$$\left|\frac{x}{|x|} - \frac{y}{|y|}\right| \leq \frac{2|y||x-y|}{|x||y|} = \frac{2|x-y|}{|x|}.$$

Multiplying both sides by $|x| = \max(|x|,|y|)$, we deduce

$$\max(|x|,|y|)\left|\frac{x}{|x|} - \frac{y}{|y|}\right| \leq 2|x-y|.$$

The result follows.

Exercise 3

Problem 3. Show that:

(a) For all $x \in \mathbb{R}$, $E(x+1) = E(x) + 1$

(b) For all $(x,y) \in \mathbb{R}^2$, $x \leq y \Rightarrow E(x) \leq E(y)$

(c) For all $x \in \mathbb{R}$, $0 \leq E(2x) - 2E(x) \leq 1$

(d) For every $n \in \mathbb{Z}^+$ and $x \in \mathbb{R}$, $0 \leq E(nx) - nE(x) \leq n-1$

Solution. **(a)** Let $x \in \mathbb{R}$. There exists a unique integer k such that:

$$k \leq x < k+1$$

Then $E(x) = k$. For $x+1$:

$$k+1 \leq x+1 < (k+1)+1$$

Thus:

$$E(x+1) = k+1 = E(x)+1$$

(b) For any $x \leq y$, let:

$$k = E(x) = \lfloor x \rfloor$$

$$m = E(y) = \lfloor y \rfloor$$

Since $x \leq y$, we have:

$$k \leq x \leq y < m+1$$

This implies $k \leq m$ because: - If $k > m$, then $m \leq k \leq x \leq y < m+1$ would force $k = m$, a contradiction Thus $E(x) = k \leq m = E(y)$

(c) Let $x = k + r$ where $k \in \mathbb{Z}$ and $r \in [0,1)$. Then:

$$E(x) = k$$

$$E(2x) = E(2k+2r) = 2k + E(2r)$$

Now analyze $E(2r)$:

- When $0 \leq r < 0.5$, $E(2r) = 0$
- When $0.5 \leq r < 1$, $E(2r) = 1$

Thus:

$$E(2x) - 2E(x) = E(2r) \in \{0, 1\}$$

proving $0 \leq E(2x) - 2E(x) \leq 1$

(d) Generalizing part (c), write $x = k + r$ with $k \in \mathbb{Z}$, $r \in [0, 1)$:

$$E(x) = k$$

$$E(nx) = E(nk + nr) = nk + E(nr)$$

Thus:

$$E(nx) - nE(x) = E(nr)$$

Since $nr \in [0, n)$, the floor function gives:

$$0 \leq E(nr) \leq n - 1$$

The maximum occurs when r approaches 1 (but $r < 1$), making $E(nr) = n - 1$

Exercise 4

Problem 4. For every positive integer n and real number x, prove:

(a) $\left\lfloor \frac{\lfloor nx \rfloor}{n} \right\rfloor = \lfloor x \rfloor$

(b) $\sum_{k=0}^{n-1} \left\lfloor x + \frac{k}{n} \right\rfloor = \lfloor nx \rfloor$

Solution. Part (a):

Let $m = \lfloor x \rfloor$ and $r = x - m$ where $0 \leq r < 1$.

1. *Lower Bound:*

$$x \geq m \Rightarrow nx \geq nm \Rightarrow \lfloor nx \rfloor \geq nm$$
$$\Rightarrow \frac{\lfloor nx \rfloor}{n} \geq m$$
$$\Rightarrow \left\lfloor \frac{\lfloor nx \rfloor}{n} \right\rfloor \geq m$$

2. *Upper Bound:*

$$x < m + 1 \Rightarrow nx < nm + n$$
$$\Rightarrow \lfloor nx \rfloor \leq nm + n - 1$$
$$\Rightarrow \frac{\lfloor nx \rfloor}{n} \leq m + \frac{n-1}{n} < m + 1$$
$$\Rightarrow \left\lfloor \frac{\lfloor nx \rfloor}{n} \right\rfloor \leq m$$

3. *Conclusion:* From both bounds we get:

$$m \leq \left\lfloor \frac{\lfloor nx \rfloor}{n} \right\rfloor \leq m$$

Thus equality holds.

Part (b):
First Proof (Direct Calculation):
Let $x = m + r$ where $m \in \mathbb{Z}$ and $0 \leq r < 1$.

1. *Decompose the Sum:*

$$\sum_{k=0}^{n-1}\left\lfloor x+\frac{k}{n}\right\rfloor=\sum_{k=0}^{n-1}\left(m+\left\lfloor r+\frac{k}{n}\right\rfloor\right)=nm+\sum_{k=0}^{n-1}\left\lfloor r+\frac{k}{n}\right\rfloor$$

2. *Count Fractional Terms:* For each k, we have:

$$\left\lfloor r+\frac{k}{n}\right\rfloor=\begin{cases}0 & \text{if } r+\frac{k}{n}<1\\ 1 & \text{if } r+\frac{k}{n}\geq 1\end{cases}$$

   Let t be the smallest integer such that $r+\frac{t}{n}\geq 1$. Then:

$$\sum_{k=0}^{n-1}\left\lfloor r+\frac{k}{n}\right\rfloor=(n-t)$$

3. *Relate to* $\lfloor nx\rfloor$*:*

$$\lfloor nx\rfloor=\lfloor nm+nr\rfloor=nm+\lfloor nr\rfloor$$

   But $\lfloor nr\rfloor=n-t$ because:

$$t=\lceil n(1-r)\rceil\Rightarrow n-t=\lfloor nr\rfloor$$

4. *Final Equality:* Thus:

$$\sum_{k=0}^{n-1}\left\lfloor x+\frac{k}{n}\right\rfloor=nm+(n-t)=\lfloor nx\rfloor$$

Second Proof (Using Periodicity):
Define $f(x)=\sum_{k=0}^{n-1}\left\lfloor x+\frac{k}{n}\right\rfloor-\lfloor nx\rfloor$.

1. *Periodicity:*

$$\begin{aligned}f\left(x+\frac{1}{n}\right)&=\sum_{k=0}^{n-1}\left\lfloor x+\frac{k+1}{n}\right\rfloor-\lfloor nx+1\rfloor\\&=\sum_{k=1}^{n}\left\lfloor x+\frac{k}{n}\right\rfloor-(\lfloor nx\rfloor+1)\\&=f(x)+\lfloor x+1\rfloor-\lfloor x\rfloor-1=f(x)\end{aligned}$$

2. *Constant on Intervals:* On any interval $\left[\frac{m}{n},\frac{m+1}{n}\right)$:
   - $\lfloor nx\rfloor$ is constant
   - Only one term in the sum changes when x crosses $\frac{m}{n}$

3. *Initial Verification:* For $x\in[0,\frac{1}{n})$:

$$f(x)=\lfloor x\rfloor+\sum_{k=1}^{n-1}\lfloor x+\frac{k}{n}\rfloor-\lfloor nx\rfloor=0+0-0=0$$

4. *Conclusion:* Since f is periodic with period $1/n$ and identically zero on $[0,1/n)$, it is zero everywhere.

Exercise 5

Problem 5. Solve in $\mathbb{R}$ the equation:

$$\left\lfloor \frac{x+1}{2} \right\rfloor + \left\lfloor \frac{x+1}{4} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor = \lfloor x \rfloor + 1$$

Solution. Set

$$Z(x) = \left\lfloor \frac{x+1}{2} \right\rfloor + \left\lfloor \frac{x+1}{4} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor$$

For $x \in \mathbb{R}$ we distinguish four cases.

Case 1. $\lfloor x \rfloor = 4p$ with $p \in \mathbb{Z}$. Equivalently we have $4p \le x < 4p+1$. Therefore,

$$\begin{aligned} 2p + \frac{1}{2} \le \frac{x+1}{2} &< 2p+1, \\ p + \frac{1}{4} \le \frac{x+1}{4} &< p + \frac{1}{2}, \\ p + \frac{1}{2} \le \frac{x+2}{4} &< p + \frac{3}{4}, \end{aligned}$$

leading to

$$Z(x) = 2p + p + p = 4p = \lfloor x \rfloor \neq \lfloor x \rfloor + 1$$

Case 2. $\lfloor x \rfloor = 4p+1$ with $p \in \mathbb{Z}$. Equivalently we have $4p+1 \le x < 4p+2$. Therefore,

$$\begin{aligned} 2p + 1 \le \frac{x+1}{2} &< 2p + \frac{3}{2}, \\ p + \frac{1}{2} \le \frac{x+1}{4} &< p + \frac{3}{4}, \\ p + \frac{3}{4} \le \frac{x+2}{4} &< p + 1, \end{aligned}$$

leading to

$$Z(x) = 2p + 1 + p + p = 4p + 1 = \lfloor x \rfloor \neq \lfloor x \rfloor + 1$$

Case 3. $\lfloor x \rfloor = 4p+2$ with $p \in \mathbb{Z}$. Equivalently we have $4p+2 \le x < 4p+3$. Therefore,

$$\begin{aligned} 2p + \frac{3}{2} \le \frac{x+1}{2} &< 2p + 2, \\ p + \frac{3}{4} \le \frac{x+1}{4} &< p + 1, \\ p + 1 \le \frac{x+2}{4} &< p + \frac{5}{4}, \end{aligned}$$

leading to

$$Z(x) = 2p + 1 + p + (p+1) = 4p + 2 = \lfloor x \rfloor \neq \lfloor x \rfloor + 1$$

Case 4. $\lfloor x \rfloor = 4p+3$ with $p \in \mathbb{Z}$. Equivalently we have $4p+3 \le x < 4p+4$. Therefore,

$$\begin{aligned} 2p + 2 \le \frac{x+1}{2} &< 2p + \frac{5}{2}, \\ p + 1 \le \frac{x+1}{4} &< p + \frac{5}{4}, \\ p + \frac{5}{4} \le \frac{x+2}{4} &< p + \frac{6}{4}, \end{aligned}$$

leading to

$$Z(x) = 2p + 2 + (p+1) + (p+1) = 4p + 4 = \lfloor x \rfloor + 1$$

Thus the set of solutions of $Z(x) = \lfloor x \rfloor + 1$ is $\bigcup_{p \in \mathbb{Z}} [4p+3, 4p+4)$.

Exercise 6

Problem 6. Let $n \in \mathbb{N}$ and $(x_1, x_2, \ldots, x_n) \in [-1,1]^n$ such that $x_1 + x_2 + \cdots + x_n = 0$. Prove that

$$\left|\sum_{k=1}^{n} kx_k\right| \leq E\left(\frac{n^2}{4}\right).$$

Solution. First Method. Remark that

$$\sum_{k=1}^{n} kx_k = (x_1 + x_2 + \cdots + x_n) + (x_2 + \cdots + x_n) + \cdots + (x_{n-1} + x_n) + x_n.$$

Since $\sum_{k=1}^{n} x_k = 0$, we have $\sum_{k=i}^{n} x_k = -\sum_{k=1}^{i-1} x_k$ for each i.

Case 1. If $n = 2p$, then $\frac{n^2}{4} = p^2$ and $E\left(\frac{n^2}{4}\right) = p^2$. We have:

$$\begin{aligned}
\left|\sum_{k=1}^{n} kx_k\right| &\leq |x_1 + \cdots + x_{2p}| + |x_2 + \cdots + x_{2p}| + \cdots + |x_p + \cdots + x_{2p}| \\
&\quad + |x_{p+1} + \cdots + x_{2p}| + \cdots + |x_{2p-1} + x_{2p}| + |x_{2p}| \\
&= 0 + |-x_1| + |-x_1 - x_2| + \cdots + |-\sum_{k=1}^{p-1} x_k| \\
&\quad + |\sum_{k=p+1}^{2p} x_k| + \cdots + |x_{2p}| \\
&\leq 0 + 1 + 2 + \cdots + (p-1) + p + (p-1) + \cdots + 1 \\
&= 2 \cdot \frac{p(p-1)}{2} + p = p^2 = E\left(\frac{n^2}{4}\right).
\end{aligned}$$

Case 2. If $n = 2p+1$, then $\frac{n^2}{4} = p^2 + p + \frac{1}{4}$ and $E\left(\frac{n^2}{4}\right) = p^2 + p$. We have:

$$\begin{aligned}
\left|\sum_{k=1}^{n} kx_k\right| &\leq |x_1 + \cdots + x_{2p+1}| + \cdots + |x_{p+1} + \cdots + x_{2p+1}| \\
&\quad + |x_{p+2} + \cdots + x_{2p+1}| + \cdots + |x_{2p+1}| \\
&= 0 + |-x_1| + \cdots + |-\sum_{k=1}^{p} x_k| + |\sum_{k=p+2}^{2p+1} x_k| + \cdots + |x_{2p+1}| \\
&\leq 0 + 1 + \cdots + p + p + (p-1) + \cdots + 1 \\
&= 2 \cdot \frac{p(p+1)}{2} = p^2 + p = E\left(\frac{n^2}{4}\right).
\end{aligned}$$

Second Method. Note that $E\left(\frac{n^2}{4}\right) = \begin{cases} p^2 & \text{if } n = 2p \\ p^2 + p & \text{if } n = 2p+1 \end{cases}$

Case 1. For $n = 2p$:

$$\begin{aligned}
\left|\sum_{k=1}^{n} kx_k\right| &= \left|\sum_{k=1}^{2p} kx_k - p\sum_{k=1}^{2p} x_k\right| = \left|\sum_{k=1}^{2p} (k-p)x_k\right| \\
&\leq \sum_{k=1}^{2p} |k-p| \cdot |x_k| \leq \sum_{k=1}^{2p} |k-p| \\
&= \sum_{k=1}^{p-1} (p-k) + \sum_{k=p+1}^{2p} (k-p) = 2\sum_{k=1}^{p-1} k = p^2 - p \leq p^2 = E\left(\frac{n^2}{4}\right).
\end{aligned}$$

Case 2. For $n = 2p + 1$:

$$\begin{aligned}
\left|\sum_{k=1}^{n} kx_k\right| &= \left|\sum_{k=1}^{2p+1} kx_k - (p+1)\sum_{k=1}^{2p+1} x_k\right| = \left|\sum_{k=1}^{2p+1}(k-p-1)x_k\right| \\
&\leq \sum_{k=1}^{2p+1} |k-p-1| \cdot |x_k| \leq \sum_{k=1}^{2p+1} |k-p-1| \\
&= \sum_{k=1}^{p}(p+1-k) + \sum_{k=p+2}^{2p+1}(k-p-1) = 2\sum_{k=1}^{p} k = p^2 + p = E\left(\frac{n^2}{4}\right).
\end{aligned}$$

Thus, for all $n \in \mathbb{N}$, we have $|\sum_{k=1}^{n} kx_k| \leq E\left(\frac{n^2}{4}\right)$.

Exercise 7

Problem 7. Calculate the sum:

$$S = \sum_{k=1}^{100} \lfloor \sqrt{k} \rfloor$$

Solution. Let us decompose the sum by grouping terms where $\lfloor \sqrt{k} \rfloor$ takes constant integer values. For each integer $m \geq 0$, the condition $m \leq \sqrt{k} < m + 1$ is equivalent to $k \in [m^2, (m+1)^2)$.

For each $m \in \mathbb{N}$, the number of integers k satisfying $m^2 \leq k < (m+1)^2$ is:

$$(m+1)^2 - m^2 = 2m + 1$$

Each such k contributes m to the sum S.

We need to find the maximal M such that $(M+1)^2 \leq 101$ (since our upper bound is 100). We find:

$$M = \lfloor \sqrt{100} \rfloor = 10$$

Thus, we can decompose the sum as:

$$S = \sum_{m=1}^{10} m \cdot (2m+1) - \text{correction for } k > 100$$

However, we must account for the partial interval when $m = 10$:

- For $m = 1$ ($1 \leq k < 4$): 3 terms ($k = 1, 2, 3$)
- For $m = 2$ ($4 \leq k < 9$): 5 terms ($k = 4, \ldots, 8$)
- $\vdots$
- For $m = 10$ ($100 \leq k < 121$): but only $k = 100$ contributes since our upper limit is 100

Therefore, the exact decomposition is:

$$\begin{aligned}
S &= \sum_{m=1}^{9} m \cdot (2m+1) + 10 \cdot 1 \\
&= \sum_{m=1}^{9} (2m^2 + m) + 10 \\
&= 2\sum_{m=1}^{9} m^2 + \sum_{m=1}^{9} m + 10
\end{aligned}$$

Using the known formulas:

$$\sum_{m=1}^{n} m = \frac{n(n+1)}{2}, \quad \sum_{m=1}^{n} m^2 = \frac{n(n+1)(2n+1)}{6}$$

We compute:

$$\sum_{m=1}^{9} m = \frac{9 \cdot 10}{2} = 45$$
$$\sum_{m=1}^{9} m^2 = \frac{9 \cdot 10 \cdot 19}{6} = 285$$

Thus:

$$S = 2 \cdot 285 + 45 + 10 = 570 + 45 + 10 = 625$$

Verification: The sum can also be computed directly:

$$S = \sum_{k=1}^{3} 1 + \sum_{k=4}^{8} 2 + \cdots + \sum_{k=81}^{99} 9 + 10 = 3 \cdot 1 + 5 \cdot 2 + \cdots + 19 \cdot 9 + 1 \cdot 10 = 625$$

Exercise 8

Problem 8. For any $a, b \in \mathbb{R}$, express:

(i) $\sup\{a, b\} - \inf\{a, b\}$

(ii) $\sup\{a, b\} + \inf\{a, b\}$

in terms of a and b. Deduce expressions for $\sup\{a, b\}$ and $\inf\{a, b\}$. What about $\sup\{-a, -b\}$ and $\inf\{-a, -b\}$?

Solution. Part 1: Expressions for sup and inf

For any two real numbers a and b, we can express their maximum and minimum using absolute value:

$$\sup\{a, b\} = \frac{a + b + |a - b|}{2}$$
$$\inf\{a, b\} = \frac{a + b - |a - b|}{2}$$

Proof: Consider the two cases:

- When $a \geq b$:

$$\frac{a + b + (a - b)}{2} = a$$
$$\frac{a + b - (a - b)}{2} = b$$

- When $b > a$:

$$\frac{a + b + (b - a)}{2} = b$$
$$\frac{a + b - (b - a)}{2} = a$$

Part 2: Difference and Sum
(i) The difference:

$$\sup\{a,b\} - \inf\{a,b\} = \frac{a+b+|a-b|}{2} - \frac{a+b-|a-b|}{2} = |a-b|$$

(ii) The sum:

$$\sup\{a,b\} + \inf\{a,b\} = \frac{a+b+|a-b|}{2} + \frac{a+b-|a-b|}{2} = a+b$$

Part 3: Behavior under negation
We prove the relations for negated numbers:
1. For $\sup\{-a,-b\}$:

$$\begin{aligned} \sup\{-a,-b\} &= \frac{-a-b+|-a-(-b)|}{2} \\ &= \frac{-(a+b)+|b-a|}{2} \\ &= -\left(\frac{a+b-|a-b|}{2}\right) \\ &= -\inf\{a,b\} \end{aligned}$$

2. For $\inf\{-a,-b\}$:

$$\begin{aligned} \inf\{-a,-b\} &= \frac{-a-b-|-a-(-b)|}{2} \\ &= \frac{-(a+b)-|b-a|}{2} \\ &= -\left(\frac{a+b+|a-b|}{2}\right) \\ &= -\sup\{a,b\} \end{aligned}$$

Thus we have shown:

$$\sup\{-a,-b\} = -\inf\{a,b\}, \quad \inf\{-a,-b\} = -\sup\{a,b\}$$

Exercise 9

Problem 9. Let A be a nonempty subset of $\mathbb{R}$. Prove that:

$$\sup(-A) = -\inf(A) \quad \text{and} \quad \inf(-A) = -\sup(A)$$

where $-A = \{-x \mid x \in A\}$.

Solution. First Method (Using ϵ-Characterization)

Part 1: $\sup(-A) = -\inf(A)$
Let $m = \inf A$. We must show:

1. $-m$ is an upper bound for $-A$: For any $x \in A$, we have $m \leq x$ by definition of infimum. Thus $-x \leq -m$, showing $-m$ bounds $-A$ from above.
2. $-m$ is the least upper bound: For any $\epsilon > 0$, since $m = \inf A$, there exists $x_\epsilon \in A$ such that:

   $$x_\epsilon < m + \epsilon$$

   which implies:

   $$-x_\epsilon > -m - \epsilon$$

   Thus no number less than $-m$ can be an upper bound for $-A$.

Part 2: $\inf(-A) = -\sup(A)$
Let $M = \sup A$. We must show:

1. $-M$ is a lower bound for $-A$: For any $x \in A$, $x \le M$ by definition of supremum. Thus $-x \ge -M$, showing $-M$ bounds $-A$ from below.
2. $-M$ is the greatest lower bound: For any $\epsilon > 0$, since $M = \sup A$, there exists $y_\epsilon \in A$ such that:

$$y_\epsilon > M - \epsilon$$

which implies:

$$-y_\epsilon < -M + \epsilon$$

Thus no number greater than $-M$ can be a lower bound for $-A$.

Second Method (Using Order Properties)

Part 1: $\sup(-A) = -\inf(A)$

- For all $x \in A$, $\inf A \le x \Rightarrow -x \le -\inf A$
- If y is any upper bound for $-A$, then $\forall x \in A$, $-x \le y \Rightarrow x \ge -y$
- Thus $-y$ is a lower bound for A, so $-y \le \inf A \Rightarrow y \ge -\inf A$
- Therefore $-\inf A$ is the least upper bound of $-A$

Part 2: $\inf(-A) = -\sup(A)$

- For all $x \in A$, $x \le \sup A \Rightarrow -x \ge -\sup A$
- If z is any lower bound for $-A$, then $\forall x \in A$, $-x \ge z \Rightarrow x \le -z$
- Thus $-z$ is an upper bound for A, so $-z \ge \sup A \Rightarrow z \le -\sup A$
- Therefore $-\sup A$ is the greatest lower bound of $-A$

Both methods establish the fundamental duality:

$$\boxed{\sup(-A) = -\inf(A)} \quad \text{and} \quad \boxed{\inf(-A) = -\sup(A)}$$

Exercise 10

Problem 10. Given nonempty subsets A and B of positive real numbers, define:

$$A \cdot B := \{ab \mid a \in A, b \in B\}$$

(a) Show that $\sup(A \cdot B) = \sup A \cdot \sup B$

(b) Give an example where $\sup(A \cdot B) \ne \sup A \cdot \sup B$ for bounded sets A, B

Solution. Part (a): Proof that $\sup(A \cdot B) = \sup A \cdot \sup B$
1. **First inequality** ($\sup(A \cdot B) \le \sup A \cdot \sup B$): For any $a \in A$ and $b \in B$, we have:

$$a \le \sup A \quad \text{and} \quad b \le \sup B$$

Thus:

$$ab \le \sup A \cdot \sup B$$

This shows $\sup A \cdot \sup B$ is an upper bound for $A \cdot B$, so:

$$\sup(A \cdot B) \leq \sup A \cdot \sup B$$

2. **Second inequality ($\sup(A \cdot B) \geq \sup A \cdot \sup B$):** For any fixed $b \in B$, we have for all $a \in A$:

$$ab \leq \sup(A \cdot B) \implies a \leq \frac{\sup(A \cdot B)}{b}$$

Thus $\frac{\sup(A \cdot B)}{b}$ is an upper bound for A, so:

$$\sup A \leq \frac{\sup(A \cdot B)}{b} \implies b \leq \frac{\sup(A \cdot B)}{\sup A}$$

This holds for all $b \in B$, so:

$$\sup B \leq \frac{\sup(A \cdot B)}{\sup A} \implies \sup A \cdot \sup B \leq \sup(A \cdot B)$$

Conclusion: $\sup(A \cdot B) = \sup A \cdot \sup B$.
Part (b): Counterexample
Take $A = \{-4, -3\}$ and $B = \{-2, 1\}$. Then:

$$A \cdot B = \{8, -4, 6, -3\}$$

We have:

$$\sup A = -3, \quad \sup B = 1 \implies \sup A \cdot \sup B = -3$$

But:

$$\sup(A \cdot B) = 8 \neq -3$$

Exercise 11

Problem 11. Let A, B be nonempty subsets of $\mathbb{R}$. Define:

$$A + B := \{a + b \mid a \in A, b \in B\}$$

$$A - B := \{a - b \mid a \in A, b \in B\}$$

Show that:

(a) $\sup(A + B) = \sup A + \sup B$

(b) $\sup(A - B) = \sup A - \inf B$

Solution. Part (a): Proof that $\sup(A + B) = \sup A + \sup B$

1. **First inequality ($\sup(A + B) \leq \sup A + \sup B$):** For any $a \in A$ and $b \in B$, we have:

$$a \leq \sup A \quad \text{and} \quad b \leq \sup B$$

Thus:

$$a + b \leq \sup A + \sup B$$

This shows $\sup A + \sup B$ is an upper bound for $A + B$, so:

$$\sup(A + B) \leq \sup A + \sup B$$

2. **Second inequality ($\sup(A + B) \geq \sup A + \sup B$):** For any $a + b \in A + B$, we have:

$$a + b \leq \sup(A + B)$$

Thus for any fixed $b \in B$:

$$a \leq \sup(A+B) - b$$

This shows $\sup(A+B) - b$ is an upper bound for A, so:

$$\sup A \leq \sup(A+B) - b \implies b \leq \sup(A+B) - \sup A$$

Now this holds for all $b \in B$, so:

$$\sup B \leq \sup(A+B) - \sup A \implies \sup A + \sup B \leq \sup(A+B)$$

Conclusion: $\sup(A+B) = \sup A + \sup B$.

Part (b): Proof that $\sup(A-B) = \sup A - \inf B$

1. **First inequality** ($\sup(A-B) \leq \sup A - \inf B$)**:** For any $a \in A$ and $b \in B$, we have:

$$a \leq \sup A \quad \text{and} \quad b \geq \inf B \implies -b \leq -\inf B$$

Thus:

$$a - b \leq \sup A - \inf B$$

This shows $\sup A - \inf B$ is an upper bound for $A - B$, so:

$$\sup(A-B) \leq \sup A - \inf B$$

2. **Second inequality** ($\sup(A-B) \geq \sup A - \inf B$)**:** For any $a - b \in A - B$, we have:

$$a - b \leq \sup(A-B)$$

Thus for any fixed $b \in B$:

$$a \leq \sup(A-B) + b$$

This shows $\sup(A-B) + b$ is an upper bound for A, so:

$$\sup A \leq \sup(A-B) + b \implies b \geq \sup A - \sup(A-B)$$

Now this holds for all $b \in B$, so:

$$\inf B \geq \sup A - \sup(A-B) \implies \sup(A-B) \geq \sup A - \inf B$$

Conclusion: $\sup(A-B) = \sup A - \inf B$.

1. **For** $\inf(A+B)$:

$$\inf(A+B) = \inf A + \inf B$$

Proof:

1. For any $a + b \in A + B$:

$$a \geq \inf A \quad \text{and} \quad b \geq \inf B \implies a + b \geq \inf A + \inf B$$

   Thus $\inf A + \inf B$ is a lower bound.

2. For any $a + b \in A + B$:

$$a + b \geq \inf(A+B) \implies b \geq \inf(A+B) - a$$

   Thus $\inf(A+B) - a$ is a lower bound for B, so:

$$\inf B \geq \inf(A+B) - a \implies a \geq \inf(A+B) - \inf B$$

   Therefore $\inf(A+B) - \inf B$ is a lower bound for A, so:

$$\inf A \geq \inf(A+B) - \inf B \implies \inf(A+B) \leq \inf A + \inf B$$

2. **For** $\inf(A-B)$:

$$\inf(A-B) = \inf A - \sup B$$

Proof:

1. For any $a - b \in A - B$:

$$a \geq \inf A \quad \text{and} \quad b \leq \sup B \implies a - b \geq \inf A - \sup B$$

Thus $\inf A - \sup B$ is a lower bound.

2. For any $a - b \in A - B$:

$$a - b \geq \inf(A-B) \implies a \geq \inf(A-B) + b$$

Thus $\inf(A-B) + b$ is a lower bound for A, so:

$$\inf A \geq \inf(A-B) + b \implies b \leq \inf A - \inf(A-B)$$

Therefore $\inf A - \inf(A-B)$ is an upper bound for B, so:

$$\sup B \leq \inf A - \inf(A-B) \implies \inf(A-B) \geq \inf A - \sup B$$

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Exercise 12

Let A, B, C, D, and Ω be the subsets of $\mathbb{R}$ defined by:

$$A = \left\{ \frac{1}{2} - \frac{n}{2n+1} : n \in \mathbb{N} \cup \{0\} \right\},$$

$$B = \left\{ 1 + \frac{(-1)^n}{n} : n \in \mathbb{N} \right\},$$

$$C = \left\{ \frac{1}{n} + (-1)^n : n \in \mathbb{N} \right\},$$

$$D = \left\{ \frac{1}{2} + \frac{n}{2n+1},\ 2 - \frac{1}{n+2} : n \in \mathbb{N}_0 = \mathbb{N} \cup \{0\} \right\},$$

$$\Omega = \left\{ \frac{m+n+2mn}{m+n+mn+1} : m, n \in \mathbb{N}_0 \right\}.$$

Problem. Prove that these sets are bounded and determine (using the formal characterization) their bounds: infimum, supremum, minimum, and maximum where they exist.

Solution for A

Consider the set

$$A = \left\{ \frac{1}{2} - \frac{n}{2n+1} : n \in \mathbb{N}_0 \right\}.$$

We first establish the boundedness of A. For any $n \in \mathbb{N}_0$,

$$\frac{1}{2} - \frac{n}{2n+1} \leq \frac{1}{2},$$

since $\frac{n}{2n+1} \geq 0$. Thus, $\frac{1}{2}$ is an upper bound of A. Moreover,

$$\frac{1}{2} - \frac{n}{2n+1} \geq 0,$$

since
$$\frac{1}{2} - \frac{n}{2n+1} \geq 0 \iff \frac{1}{2}(2n+1) \geq n \iff n + \tfrac{1}{2} \geq n,$$
which is true for all $n \in \mathbb{N}_0$. Hence, 0 is a lower bound of A. Therefore, A is bounded.

We now determine the least upper bound. Clearly, $\frac{1}{2} \in A$ since for $n = 0$,
$$\frac{1}{2} - \frac{0}{1} = \frac{1}{2}.$$
Thus, $\frac{1}{2}$ is an upper bound of A and belongs to A. By definition, it follows that
$$\sup A = \frac{1}{2} \quad \text{and} \quad \max A = \frac{1}{2}.$$

Next, we determine the greatest lower bound. We claim $\inf A = 0$. We first verify that 0 is a lower bound of A, as shown above. To confirm that 0 is the greatest lower bound, we appeal to the formal characterization of the infimum.

Let $\varepsilon > 0$. We seek $a_\varepsilon \in A$ such that
$$0 \leq a_\varepsilon < \varepsilon.$$

We know that elements of A have the form
$$a = \frac{1}{2} - \frac{n}{2n+1}.$$
We impose the condition
$$\frac{1}{2} - \frac{n}{2n+1} < \varepsilon,$$
and solve for n.
$$\frac{1}{2} - \varepsilon < \frac{n}{2n+1}.$$
Multiply both sides by $2n+1$:
$$\left(\frac{1}{2} - \varepsilon\right)(2n+1) < n.$$
Expand:
$$(1-2\varepsilon)n + \frac{1}{2} - \varepsilon < n.$$
Bring n terms together:
$$(1-2\varepsilon)n - n < \varepsilon - \tfrac{1}{2}.$$
$$(-2\varepsilon)n < \varepsilon - \tfrac{1}{2}.$$

Since $-2\varepsilon < 0$, dividing both sides reverses the inequality:
$$n > \frac{\frac{1}{2} - \varepsilon}{2\varepsilon}.$$

Define:
$$n_\varepsilon = \begin{cases} 0, & \text{if } \varepsilon \geq \frac{1}{2}, \\ \left\lceil \dfrac{\frac{1}{2} - \varepsilon}{2\varepsilon} \right\rceil, & \text{if } 0 < \varepsilon < \frac{1}{2}. \end{cases}$$

Verification:

- If $\varepsilon \geq \frac{1}{2}$, take $n_\varepsilon = 0$. Then
$$a_\varepsilon = \frac{1}{2} - \frac{0}{1} = \frac{1}{2} < \varepsilon.$$

- If $0 < \varepsilon < \frac{1}{2}$, consider
$$n_\varepsilon = \left\lceil \frac{\frac{1}{2} - \varepsilon}{2\varepsilon} \right\rceil.$$
By construction,
$$\frac{1}{2} - \frac{n_\varepsilon}{2n_\varepsilon + 1} < \varepsilon.$$

Thus, for every $\varepsilon > 0$, there exists $a_\varepsilon \in A$ such that $a_\varepsilon < \varepsilon$. By the characterization of the infimum, we conclude
$$\inf A = 0.$$

Finally, note that $0 \notin A$, since for every $n \in \mathbb{N}_0$,
$$\frac{1}{2} - \frac{n}{2n+1} > 0.$$
Thus, A has no minimum.

Conclusion

$$\boxed{\inf A = 0,\ \sup A = \tfrac{1}{2},\ \max A = \tfrac{1}{2},\ \min A \text{ does not exist}}$$

Solution for B

Consider the set
$$B = \left\{ 1 + \frac{(-1)^n}{n} : n \in \mathbb{N} \right\}.$$

Boundedness. For all $n \in \mathbb{N}$, we have:
$$1 + \frac{(-1)^n}{n} \leq 1 + \frac{1}{n} \leq 2,$$
$$1 + \frac{(-1)^n}{n} \geq 1 - \frac{1}{n} \geq 0.$$
Therefore, B is bounded above by 2 and below by 0.

Determination of the supremum. We claim that $\sup B = \frac{3}{2}$. Observe that for $n = 2$, we have:
$$1 + \frac{(-1)^2}{2} = 1 + \frac{1}{2} = \frac{3}{2} \in B.$$
For any even $n = 2k$, we have:
$$1 + \frac{(-1)^{2k}}{2k} = 1 + \frac{1}{2k} < \frac{3}{2} \quad \text{for } k \geq 2.$$
For any odd $n = 2k+1$, we have:
$$1 + \frac{(-1)^{2k+1}}{2k+1} = 1 - \frac{1}{2k+1} < 1 < \frac{3}{2}.$$
Therefore, no element of B exceeds $\frac{3}{2}$, and since $\frac{3}{2} \in B$, it follows that:
$$\sup B = \frac{3}{2}, \quad \max B = \frac{3}{2}.$$

Characterization of the supremum. Let $\varepsilon > 0$. We seek $b_\varepsilon \in B$ such that:
$$\frac{3}{2} - \varepsilon < b_\varepsilon \leq \frac{3}{2}.$$

We consider even $n = 2k$, for $k \in \mathbb{N}$, so:

$$b_\varepsilon = 1 + \frac{1}{2k}.$$

We require:

$$1 + \frac{1}{2k} > \frac{3}{2} - \varepsilon.$$

Subtracting 1 from both sides:

$$\frac{1}{2k} > \frac{1}{2} - \varepsilon.$$

Since $\frac{1}{2} - \varepsilon > 0$ for $\varepsilon < \frac{1}{2}$, we solve:

$$2k < \frac{1}{\frac{1}{2} - \varepsilon} \quad \Longrightarrow \quad k < \frac{1}{2\left(\frac{1}{2} - \varepsilon\right)}.$$

Choose any natural number k satisfying this inequality. Then $n = 2k$, and:

$$b_\varepsilon = 1 + \frac{1}{n} \in B,$$

and

$$\frac{3}{2} - \varepsilon < b_\varepsilon \leq \frac{3}{2}.$$

If $\varepsilon \geq \frac{1}{2}$, we simply take $b_\varepsilon = \frac{3}{2} \in B$, and the condition is clearly satisfied.

Determination of the infimum. We claim that $\inf B = 0$. For $n = 1$, we have:

$$1 + \frac{(-1)^1}{1} = 1 - 1 = 0 \in B.$$

For any $n \in \mathbb{N}$, we know:

$$1 + \frac{(-1)^n}{n} \geq 0.$$

Therefore, 0 is a lower bound of B, and since it is attained, we have:

$$\inf B = 0, \quad \min B = 0.$$

Characterization of the infimum. Let $\varepsilon > 0$. We seek $b_\varepsilon \in B$ such that:

$$0 \leq b_\varepsilon < \varepsilon.$$

Take $b_\varepsilon = 0 \in B$, then clearly $b_\varepsilon < \varepsilon$ for any $\varepsilon > 0$, so the condition is satisfied.

Conclusion

$$\boxed{\inf B = 0, \quad \sup B = \tfrac{3}{2}, \quad \min B = 0, \quad \max B = \tfrac{3}{2}}$$

Solution for C

Consider the set

$$C = \left\{ \frac{1}{n} + (-1)^n : n \in \mathbb{N} \right\}.$$

Boundedness. For all $n \in \mathbb{N}$, note that $(-1)^n = \pm 1$. Consider two cases:
- If n is even, $n = 2k$, then

$$\frac{1}{n} + (-1)^n = \frac{1}{2k} + 1.$$

Thus:

$$\frac{1}{2k} + 1 \leq 1 + \frac{1}{2} = \frac{3}{2}.$$

- If n is odd, $n = 2k+1$, then

$$\frac{1}{n} + (-1)^n = \frac{1}{2k+1} - 1.$$

Thus:

$$\frac{1}{2k+1} - 1 < 0.$$

In both cases, we have:

$$\frac{1}{n} + (-1)^n < \frac{3}{2} \quad \text{and} \quad \frac{1}{n} + (-1)^n > -1.$$

Hence C is bounded above by $\frac{3}{2}$ and below by -1.

—

Determination of the supremum. We claim that

$$\sup C = \frac{3}{2}.$$

For $n = 2$,

$$\frac{1}{2} + (-1)^2 = \frac{1}{2} + 1 = \frac{3}{2} \in C.$$

For even $n \geq 2$,

$$\frac{1}{2k} + 1 < \frac{3}{2}.$$

For odd n,

$$\frac{1}{2k+1} - 1 < 1 < \frac{3}{2}.$$

Thus no element of C exceeds $\frac{3}{2}$, and since $\frac{3}{2} \in C$, we have:

$$\sup C = \frac{3}{2}, \quad \max C = \frac{3}{2}.$$

—

Characterization of $\sup C$. Let $\varepsilon > 0$. We seek $c_\varepsilon \in C$ such that:

$$\frac{3}{2} - \varepsilon < c_\varepsilon \leq \frac{3}{2}.$$

Consider even $n = 2k$. Then:

$$c_\varepsilon = \frac{1}{2k} + 1.$$

We require:

$$\frac{1}{2k} + 1 > \frac{3}{2} - \varepsilon.$$

Subtracting 1:

$$\frac{1}{2k} > \frac{1}{2} - \varepsilon.$$

Since $\frac{1}{2} - \varepsilon > 0$ for $\varepsilon < \frac{1}{2}$, solve:

$$2k < \frac{1}{\frac{1}{2} - \varepsilon}.$$

Take any $k \in \mathbb{N}$ satisfying this inequality. Then $n = 2k$ and

$$c_\varepsilon = \frac{1}{n} + 1 \in C,$$

with

$$\frac{3}{2} - \varepsilon < c_\varepsilon \leq \frac{3}{2}.$$

If $\varepsilon \geq \frac{1}{2}$, take $c_\varepsilon = \frac{3}{2} \in C$.

—

Determination of the infimum. We claim:

$$\inf C = -1.$$

For odd $n = 2k+1$,

$$\frac{1}{2k+1} + (-1)^{2k+1} = \frac{1}{2k+1} - 1.$$

As $k \to \infty$,

$$\frac{1}{2k+1} - 1 \to -1.$$

Moreover, for $k = 0$,

$$\frac{1}{1} - 1 = 0 \in C.$$

For even n,

$$\frac{1}{2k} + 1 > 0 > -1.$$

Thus -1 is a lower bound of C. For any $\delta > 0$, we can find k such that:

$$\frac{1}{2k+1} - 1 > -1 - \delta,$$

but -1 is never attained. Hence:

$$\inf C = -1, \quad \min C \text{ does not exist.}$$

—

Characterization of $\inf C$. Let $\varepsilon > 0$. We seek $c_\varepsilon \in C$ such that:

$$-1 \leq c_\varepsilon < -1 + \varepsilon.$$

Consider odd $n = 2k+1$. Then:

$$c_\varepsilon = \frac{1}{2k+1} - 1.$$

We require:

$$\frac{1}{2k+1} - 1 < -1 + \varepsilon.$$

Adding 1 to both sides:

$$\frac{1}{2k+1} < \varepsilon.$$

Solve:

$$2k+1 > \frac{1}{\varepsilon}.$$

Take any $k \in \mathbb{N}_0$ satisfying this inequality. Then $n = 2k+1$ and:

$$c_\varepsilon = \frac{1}{n} - 1 \in C,$$

with:

$$-1 \leq c_\varepsilon < -1 + \varepsilon.$$

—

Conclusion

$$\boxed{\inf C = -1,\ \sup C = \tfrac{3}{2},\ \min C \text{ does not exist},\ \max C = \tfrac{3}{2}}$$

Solution for D

Consider the set

$$D = \left\{ \frac{1}{2} + \frac{n}{2n+1},\ 2 - \frac{1}{n+2} : n \in \mathbb{N}_0 = \mathbb{N} \cup \{0\} \right\}.$$

We write D as the union of two subsets:

$$D_1 = \left\{ \frac{1}{2} + \frac{n}{2n+1} : n \in \mathbb{N}_0 \right\}, \quad D_2 = \left\{ 2 - \frac{1}{n+2} : n \in \mathbb{N}_0 \right\}.$$

—

Scratch work: finding the bounds. To determine the bounds of D_1, observe that for $n \in \mathbb{N}_0$,

$$\frac{1}{2} < \frac{1}{2} + \frac{n}{2n+1} < 1.$$

Indeed, as $n \to \infty$,

$$\frac{1}{2} + \frac{n}{2n+1} \to \frac{1}{2} + \frac{1}{2} = 1.$$

For D_2, note:

$$\frac{3}{2} < 2 - \frac{1}{n+2} < 2,$$

and as $n \to \infty$,

$$2 - \frac{1}{n+2} \to 2.$$

This use of limits and monotonicity is only a **scratch work** to help find the bounds. In the formal proof below, we avoid sequences since they are not yet studied.

—

Boundedness. Since

$$\forall n \in \mathbb{N}_0,\ \frac{1}{2} < \frac{1}{2} + \frac{n}{2n+1} < 1,$$

and

$$\forall n \in \mathbb{N}_0,\ \frac{3}{2} < 2 - \frac{1}{n+2} < 2,$$

we deduce:

$$\inf D_1 = \frac{1}{2}, \quad \sup D_1 = 1,$$
$$\inf D_2 = \frac{3}{2}, \quad \sup D_2 = 2.$$

Since $D = D_1 \cup D_2$, we have:

$$\inf D = \min(\inf D_1, \inf D_2) = \min\left(\frac{1}{2}, \frac{3}{2}\right) = \frac{1}{2},$$
$$\sup D = \max(\sup D_1, \sup D_2) = \max(1, 2) = 2.$$

—

Characterization of $\sup D_1$. We claim that $\sup D_1 = 1$. Let $\varepsilon > 0$. We seek $d_{1,\varepsilon} \in D_1$ such that:

$$1 - \varepsilon < d_{1,\varepsilon} \leq 1.$$

Consider

$$d_{1,\varepsilon} = \frac{1}{2} + \frac{n}{2n+1}.$$

We require:

$$\frac{1}{2} + \frac{n}{2n+1} > 1 - \varepsilon.$$

Subtract $\frac{1}{2}$:

$$\frac{n}{2n+1} > \frac{1}{2} - \varepsilon.$$

Solve:

$$2n + 1 < \frac{n}{\frac{1}{2} - \varepsilon}.$$

For $\varepsilon < \frac{1}{2}$, this is possible for large enough n. Choose $n_\varepsilon \in \mathbb{N}_0$ satisfying:

$$n > \frac{1}{2\left(\frac{1}{2} - \varepsilon\right)}.$$

Then

$$d_{1,\varepsilon} = \frac{1}{2} + \frac{n_\varepsilon}{2n_\varepsilon + 1} \in D_1,$$

with

$$1 - \varepsilon < d_{1,\varepsilon} \leq 1.$$

—

Characterization of $\inf D_1$. We claim that $\inf D_1 = \frac{1}{2}$. Let $\varepsilon > 0$. We seek $d_{1,\varepsilon} \in D_1$ such that:

$$\frac{1}{2} \leq d_{1,\varepsilon} < \frac{1}{2} + \varepsilon.$$

For $n = 0$,

$$d_{1,\varepsilon} = \frac{1}{2} + \frac{0}{1} = \frac{1}{2} \in D_1,$$

and

$$\frac{1}{2} \leq d_{1,\varepsilon} < \frac{1}{2} + \varepsilon.$$

—

Characterization of $\sup D_2$. We claim that $\sup D_2 = 2$. Let $\varepsilon > 0$. We seek $d_{2,\varepsilon} \in D_2$ such that:

$$2 - \varepsilon < d_{2,\varepsilon} < 2.$$

Consider

$$d_{2,\varepsilon} = 2 - \frac{1}{n+2}.$$

We require:

$$2 - \frac{1}{n+2} > 2 - \varepsilon.$$

Subtract 2 and multiply by -1:

$$\frac{1}{n+2} < \varepsilon.$$

Solve:

$$n + 2 > \frac{1}{\varepsilon}.$$

Choose $n_\varepsilon \in \mathbb{N}_0$ satisfying this inequality. Then

$$d_{2,\varepsilon} = 2 - \frac{1}{n_\varepsilon + 2} \in D_2,$$

with

$$2 - \varepsilon < d_{2,\varepsilon} < 2.$$

—

Characterization of $\inf D_2$. We claim that $\inf D_2 = \frac{3}{2}$. Let $\varepsilon > 0$. We seek $d_{2,\varepsilon} \in D_2$ such that:

$$\frac{3}{2} \le d_{2,\varepsilon} < \frac{3}{2} + \varepsilon.$$

For $n = 0$,

$$d_{2,\varepsilon} = 2 - \frac{1}{2} = \frac{3}{2} \in D_2,$$

and

$$\frac{3}{2} \le d_{2,\varepsilon} < \frac{3}{2} + \varepsilon.$$

—

Conclusion

$$\boxed{\inf D = \tfrac{1}{2},\ \sup D = 2,\ \min D = \tfrac{1}{2},\ \max D \text{ does not exist}}$$

Solution for Ω

Consider the set

$$\Omega = \left\{ \frac{m+n+2mn}{m+n+mn+1} : m, n \in \mathbb{N}_0 \right\}.$$

—

Scratch work: finding the bounds. We simplify:

$$\frac{m+n+2mn}{m+n+mn+1} = \frac{(m+n)+2mn}{(m+n)+mn+1}.$$

Upper bound: For fixed $m \ge 1$, consider $n \to \infty$.

$$\frac{m+n+2mn}{m+n+mn+1} = \frac{2mn+\cdots}{mn+\cdots} \approx \frac{2m}{m} = 2.$$

Thus, $\lim_{n\to\infty} \frac{m+n+2mn}{m+n+mn+1} = 2$.

Lower bound: For $m, n \ge 0$,

$$\frac{m+n+2mn}{m+n+mn+1} > 0.$$

For $m = 0$ and $n \to \infty$,

$$\frac{n}{n+1} \to 1.$$

For $m = n = 0$,

$$\frac{0+0+0}{0+0+0+1} = \frac{0}{1} = 0 \in \Omega.$$

Hence the bounds are:

$$\boxed{\inf \Omega = 0,\ \sup \Omega = 2.}$$

We now rigorously prove these.

—

Boundedness. *Upper bound:* For $m, n \ge 0$,

$$m+n+2mn < 2(m+n+mn+1).$$

Indeed:

$$m+n+2mn < 2m+2n+2mn+2,$$

so:

$$\frac{m+n+2mn}{m+n+mn+1} < \frac{2m+2n+2mn+2}{m+n+mn+1} = 2 + \frac{2}{m+n+mn+1}.$$

Since $\frac{2}{m+n+mn+1} > 0$,

$$\frac{m+n+2mn}{m+n+mn+1} < 2.$$

Thus, 2 is an upper bound of Ω.

Lower bound: Clearly,

$$\frac{m+n+2mn}{m+n+mn+1} \geq 0.$$

Thus, 0 is a lower bound.

—

Determination of $\sup \Omega$. We claim:

$$\sup \Omega = 2.$$

2 is an upper bound of Ω, and for any $\varepsilon > 0$, we must show there exists $\omega_\varepsilon \in \Omega$ such that:

$$2 - \varepsilon < \omega_\varepsilon < 2.$$

Consider $m, n \geq 1$. For fixed m, take

$$\omega_\varepsilon = \frac{m+n+2mn}{m+n+mn+1}.$$

We want:

$$\frac{m+n+2mn}{m+n+mn+1} > 2 - \varepsilon.$$

Solve:

$$m+n+2mn > (2-\varepsilon)(m+n+mn+1).$$

Expanding:

$$m+n+2mn > (2-\varepsilon)(m+n) + (2-\varepsilon)mn + (2-\varepsilon).$$

This inequality can be satisfied for large n (details omitted for brevity). Take n_ε large enough so that:

$$\frac{m+n_\varepsilon+2mn_\varepsilon}{m+n_\varepsilon+mn_\varepsilon+1} > 2 - \varepsilon.$$

Thus, such $\omega_\varepsilon \in \Omega$ exists.

—

Determination of $\inf \Omega$. We claim:

$$\inf \Omega = 0.$$

Clearly,

$$\frac{m+n+2mn}{m+n+mn+1} \geq 0.$$

For any $\varepsilon > 0$, we must show there exists $\omega_\varepsilon \in \Omega$ such that:

$$0 \leq \omega_\varepsilon < \varepsilon.$$

Take $m = 0$, then:

$$\omega_\varepsilon = \frac{n}{n+1}.$$

We want:

$$\frac{n}{n+1} < \varepsilon.$$

Solve:

$$n < \varepsilon(n+1).$$

For $\varepsilon < 1$, this is false, so instead take $m = n = 0:$

$$\omega_\varepsilon = \frac{0}{1} = 0 \in \Omega,$$

which satisfies:

$$0 \leq 0 < \varepsilon.$$

Thus, $\inf \Omega = 0$.

—

Conclusion

$$\boxed{\inf\Omega = 0,\ \sup\Omega = 2,\ \min\Omega = 0,\ \max\Omega \text{ does not exist}}$$

Exercise 13

Show that

$$\sup\left\{x \in \mathbb{Q} : x > 0,\ x^2 < 2\right\} = \sqrt{2}.$$

—

Solution.
Define the set

$$S = \left\{x \in \mathbb{Q} : x > 0,\ x^2 < 2\right\}.$$

We claim:

$$\boxed{\sup S = \sqrt{2}}.$$

—

Step 1: $\sqrt{2}$ is an upper bound of S.
Let $x \in S$. By definition of S, $x > 0$ and $x^2 < 2$. Since $\sqrt{2} > 0$, we have:

$$x^2 < 2 = (\sqrt{2})^2.$$

Since $x, \sqrt{2} > 0$, taking square roots preserves the inequality:

$$x < \sqrt{2}.$$

Thus, every $x \in S$ satisfies $x < \sqrt{2}$. Hence, $\sqrt{2}$ is an upper bound for S.

—

Step 2: $\sqrt{2}$ is the least upper bound of S.
Let u be any upper bound of S. We must show:

$$u \geq \sqrt{2}.$$

Proof by contradiction: Suppose $u < \sqrt{2}$. Set

$$v = \frac{u + \sqrt{2}}{2}.$$

Since $u < \sqrt{2}$, we have:

$$u < v < \sqrt{2}.$$

Now compute v^2 :

$$v^2 = \left(\frac{u + \sqrt{2}}{2}\right)^2 = \frac{u^2 + 2u\sqrt{2} + 2}{4}.$$

Since $u < \sqrt{2}$, we know $u^2 < 2$. Also, $2u\sqrt{2} > 0$. Thus:

$$v^2 < \frac{2 + 2u\sqrt{2} + 2}{4} = \frac{4 + 2u\sqrt{2}}{4}.$$

But $2u\sqrt{2} < 2\sqrt{2}\sqrt{2} = 4$. Hence:

$$4 + 2u\sqrt{2} < 4 + 4 = 8,$$

so:

$$v^2 < \frac{8}{4} = 2.$$

We conclude:

$$v \in \mathbb{R},\ v < \sqrt{2},\ v^2 < 2.$$

Since $\mathbb{Q}$ is dense in $\mathbb{R}$, there exists $q \in \mathbb{Q}$ such that:

$$u < q < v.$$

(Note: here, this argument does not depend on density because $u, v \in \mathbb{R}$ and $u < v$, so rational approximations exist by Archimedean property.)

Observe:

$$q^2 < v^2 < 2,$$

so $q^2 < 2$. Also, $q > 0$. Thus $q \in S$.

But $q > u$. This contradicts the assumption that u is an upper bound of S.

—

Conclusion.

The assumption $u < \sqrt{2}$ leads to a contradiction. Therefore, any upper bound u of S satisfies $u \geq \sqrt{2}$. Since $\sqrt{2}$ is itself an upper bound, it is the least upper bound:

$$\boxed{\sup S = \sqrt{2}}.$$